

Forecasting Weekly German Electricity Demand:

A Comparative Study of Benchmark, SARIMA/SARIMAX, Feature-Based and Recurrent-Network Models

Advanced Research Topics — Time-Series Case Study (Assignment 1)

Abstract

This study forecasts the weekly average electricity load of the German transmission system (Open Power System Data, January 2015 – October 2020). The hourly series is aggregated to a weekly grid of 301 observations and the final two years (104 weeks) are held out as a genuine forecast horizon. After exploratory analysis and stationarity testing (ADF, KPSS, seasonal and first differencing, ACF/PACF), five model families are compared under identical evaluation: naive benchmarks, an AIC-selected SARIMA, a SARIMAX with exogenous temperature, feature-based tree ensembles (Random Forest, Gradient Boosting) and an LSTM. All are scored by MAE, RMSE, MASE and bias. **The central finding is methodological: compared like-for-like as true multi-step forecasts, no model meaningfully beats the seasonal-naive benchmark over two years.** Models that appear to win consume recent actual load (multi-step forecast leakage), and the elevated errors across all methods trace to the 2020 COVID-19 demand collapse, which lies inside the test window and is unforecastable from pre-2019 data.

1. Introduction

Accurate load forecasting underpins generation scheduling and reserve procurement, where errors translate directly into cost. National demand is strongly patterned — high in winter, low in summer, lower at weekends, cycling within the day — which suits classical seasonal methods, yet it is also weather- and behaviour-driven, which invites machine learning. This work asks a sharper question than which curve fits best: over a two-year horizon, can any model genuinely improve on the simplest seasonal benchmark, and where the numbers say yes, is the gain real or an artefact of how the forecast is produced? Five model families are implemented end-to-end under one protocol, with close attention to whether each makes a genuine multi-step forecast or quietly consumes information unavailable at the forecast origin.

2. Data and Exploratory Analysis

The actual German load reported to ENTSO-E is taken at hourly resolution, placed on a regular grid with gaps interpolated (50,400 hours to 30 September 2020) and resampled to weekly *average power* in GW using the mean, so units stay comparable across frequencies. The resulting 301-week series has mean 55.48 GW, standard deviation 3.76 GW and range 46.51–63.59 GW.

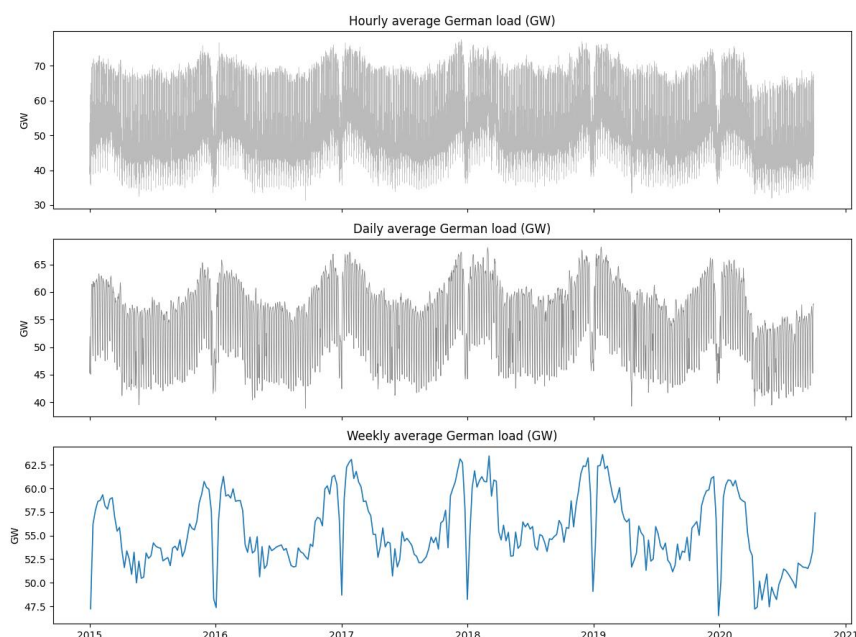


Figure 1. Load at hourly, daily and weekly resolution (GW). Aggregation removes the intraday and weekly cycles, exposing the annual wave and the 2020 COVID-19 dip.

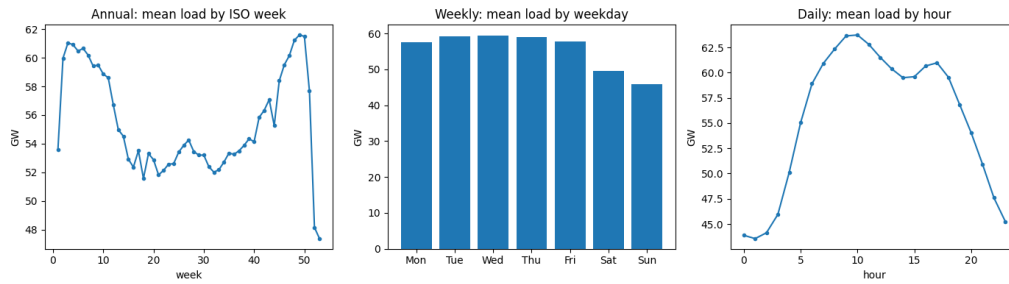


Figure 2. Seasonal profiles: mean load by ISO week (left), weekday (centre) and hour of day (right).

Figures 1–2 show the structure clearly: aggregation strips the intraday swing and weekend effect to leave a clean annual wave falling from a January maximum to an August minimum; weekdays sit well above weekends; the daily profile shows the morning ramp and overnight trough. A pronounced dip in early 2020 marks the COVID-19 lockdown — critically, it lies inside the two-year test window.

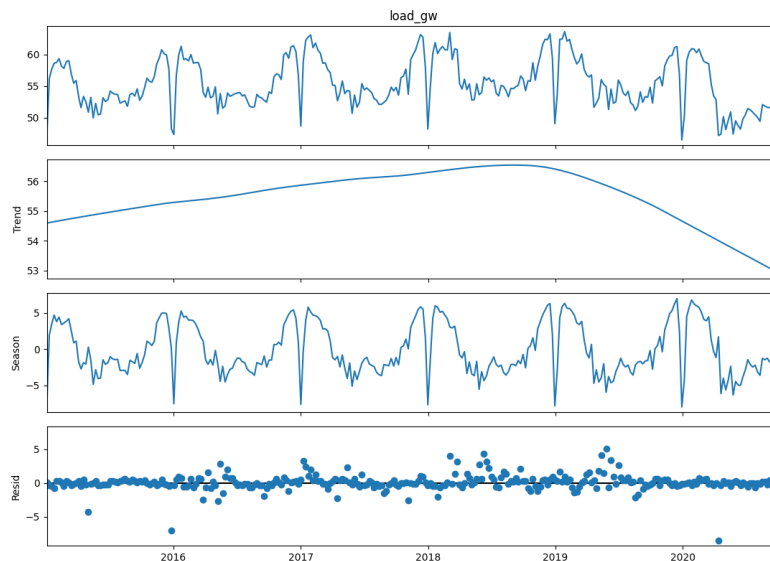


Figure 3. STL decomposition (period 52). The seasonal component is large and stable; the trend is comparatively weak.

A robust STL decomposition (Figure 3) gives a seasonal strength of 0.871 against a trend strength of only 0.383 (0–1 scale). A seasonal strength of 0.87 means a good forecast must, above all, reproduce the annual cycle — the observation that drives the whole comparison.

3. Stationarity Analysis

ARIMA-type models require stationary input, so the series was tested with the Augmented Dickey–Fuller (ADF; null = unit root) and KPSS (null = stationarity) tests in levels, after a first difference ($d=1$), a seasonal difference at lag 52 ($D=1$), and both. Using a test and its converse together is deliberate.

Transformation	ADF stat	ADF p	ADF verdict	KPSS stat	KPSS p	KPSS verdict
Levels	-4.047	0.0012	Stationary	0.160	0.100	Stationary
First diff ($d=1$)	-7.069	0.0000	Stationary	0.056	0.100	Stationary
Seasonal diff ($D=1$, lag 52)	-4.295	0.0005	Stationary	1.342	0.010	Non-stationary
$d=1 + D=1$ (lag 52)	-7.278	0.0000	Stationary	0.105	0.100	Stationary

Table 1. Stationarity tests on the weekly load series (significance level 0.05).

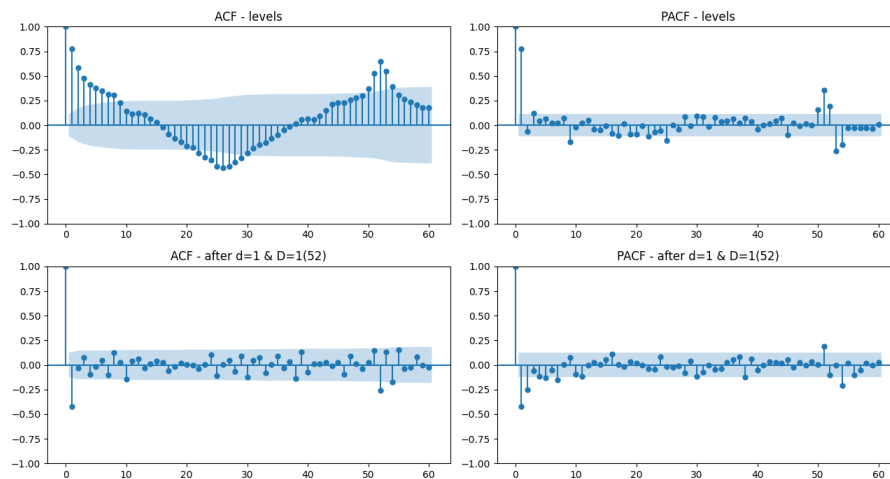


Figure 4. ACF/PACF in levels (top) and after first plus seasonal differencing (bottom). Differencing removes the slow seasonal decay, leaving short-lag structure.

In levels both tests already call the series stationary — a caution, not a green light: ADF and KPSS are effectively blind to seasonal non-stationarity, so they miss the 0.87-strength annual cycle and understate the need to difference. Applying both a first and a seasonal difference (Table 1, final row) leaves both tests stationary and collapses the ACF/PACF (Figure 4) to a few low-order spikes, which fixes the differencing order for the SARIMA search and points to low-order MA terms.

4. Methods and Rationale

Five families were chosen to span trivial to expressive, so the value added by complexity is measured, not assumed.

Benchmarks. Mean, naive, seasonal naive (the last full 52-week season repeated) and drift. The seasonal naive is a demanding competitor because it reproduces the annual cycle for free; it is the model to beat.

SARIMA / SARIMAX. SARIMA suits a linear, strongly seasonal series whose autocorrelation differencing renders stationary; orders were chosen by AIC grid search with the seasonal order fixed at (0,1,1,52). SARIMAX adds weekly Berlin temperature (Open-Meteo) with heating- and cooling-degree terms as an exogenous regressor.

Tree ensembles. Random Forest and Gradient Boosting on engineered features (load lags 1–52, rolling means, calendar terms, Fourier harmonics, temperature). Trees capture non-linear interaction and expose feature importances. Each is run both recursively (a genuine multi-step forecast) and one-step-ahead (allowed to see recent actual load) to isolate the effect of that information.

LSTM. A recurrent network on the hourly series (48-hour look-back), aggregated back to weekly. As implemented it is a walk-forward one-step-ahead forecaster — a distinction that matters for reading its score.

5. Experimental Setup

The final 104 weeks (Oct 2018 – Oct 2020) are the test horizon; the prior 197 weeks train the models. Scores are MAE, RMSE, MASE and bias. MASE is the headline: it divides forecast error by the in-sample seasonal-naive one-step error, so below 1 beats that benchmark and above 1 does not, independent of units. Decisively, the benchmark, SARIMA, SARIMAX and the recursive tree forecasts are all **genuine multi-step forecasts** made once at the origin using no test-period load — which is what makes the comparison fair and what exposes leakage in the models allowed to see recent actuals.

6. Results

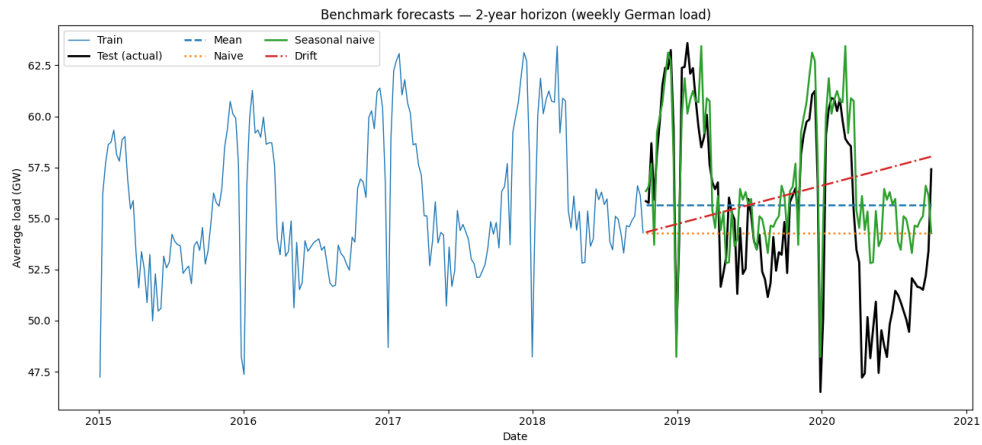


Figure 5. Benchmark forecasts over two years. Only the seasonal naive tracks the annual cycle.

Benchmark	MAE	RMSE	MASE	Bias
Seasonal naive	2.319	3.007	1.732	1.732
Naive	3.783	4.459	2.827	-0.882
Mean	3.789	4.397	2.831	0.481
Drift	4.340	5.118	3.243	1.007

Table 2. Benchmark accuracy over the two-year horizon (GW; best row highlighted).

The seasonal naive dominates the benchmarks (MASE 1.73) because it alone encodes the annual cycle; mean, naive and drift draw flat or sloped lines through a wave and are ~60% worse. That the seasonal naive still exceeds 1 is itself a signal that 2019–2020 is harder than the 2015–2018 data used to scale MASE — the test window is anomalous, not the benchmark weak.

6.1 SARIMA

The AIC search selected $(0,1,2)(0,1,1,52)$ — one non-seasonal and one seasonal difference with low-order MA terms, exactly as the differenced ACF anticipated (AIC 317.2, BIC 327.2). Ljung–Box p-values of 0.83, 0.99 and 0.78 at lags 13, 26 and 52 leave the residuals effectively white noise (Figure 6): the model captures the in-sample autocorrelation well.

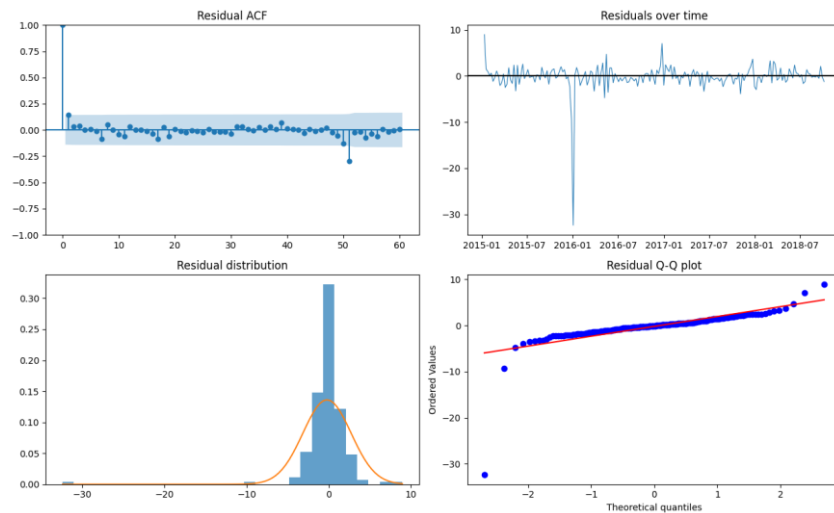


Figure 6. SARIMA residual diagnostics: residual ACF, residuals over time, distribution vs a fitted normal, and Q-Q plot.

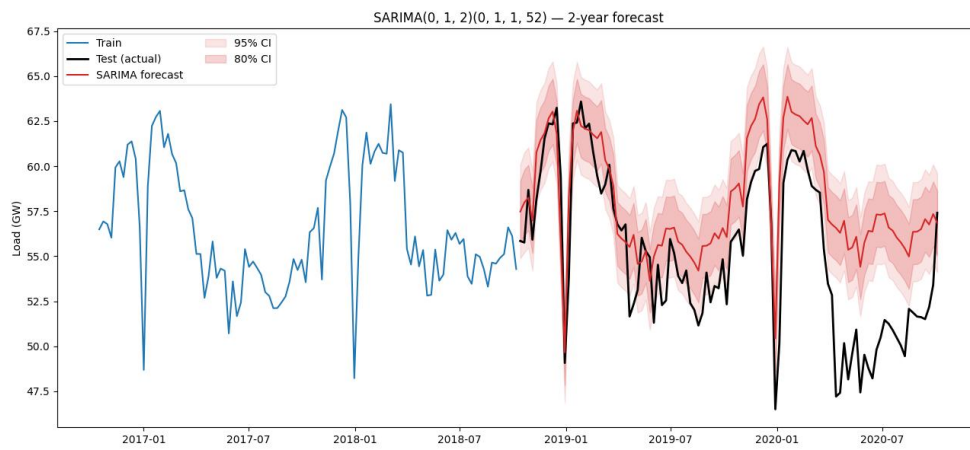


Figure 7. SARIMA $(0,1,2)(0,1,1,52)$ two-year forecast with 80% and 95% confidence intervals against the actuals.

Yet the point forecast (Figure 7) is poor: MAE 3.13, RMSE 3.83, MASE 2.34 — worse than seasonal naive. It faithfully extrapolates the pre-2019 pattern into a 2020 collapse that falls outside its bands. Good in-sample fit does not guarantee good long-horizon forecasting when the future breaks from the past.

6.2 SARIMAX with temperature

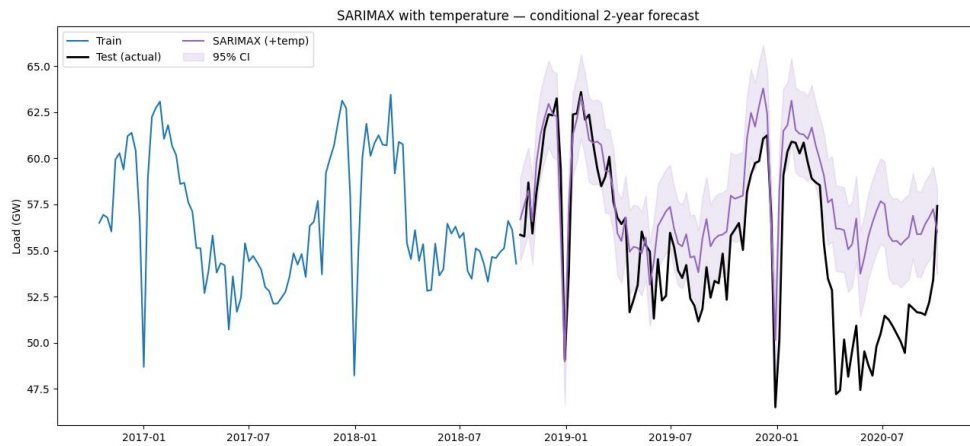


Figure 8. SARIMAX with exogenous temperature — conditional two-year forecast with 95% confidence interval.

Temperature helps modestly (RMSE 3.59, MASE 2.13) with a positive coefficient confirming heating demand, but it still does not beat the seasonal naive, and this is a conditional forecast handed the true future temperature (see Q4).

6.3 Feature-based ensembles and LSTM

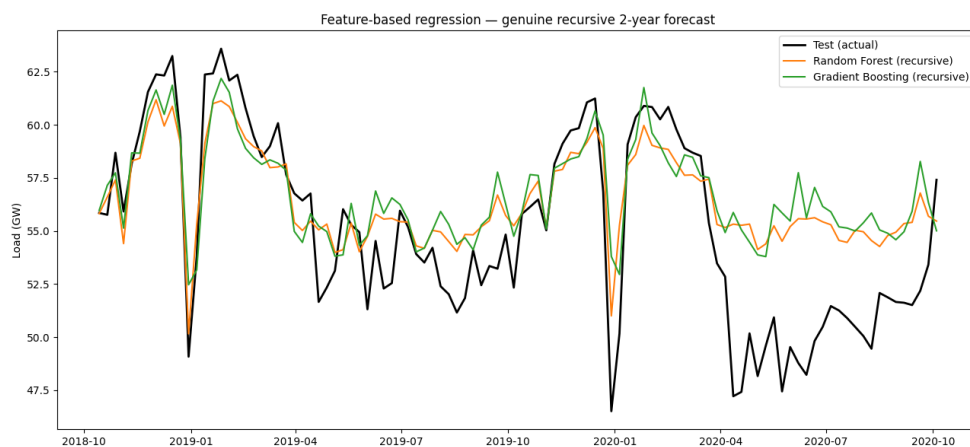


Figure 9. Genuine recursive two-year forecasts from the Random Forest and Gradient Boosting regressors.

Model	MAE	RMSE	MASE	Bias
Random Forest (recursive)	2.383	3.018	1.781	1.274
Gradient Boosting (recursive)	2.531	3.317	1.891	1.561

Model	MAE	RMSE	MASE	Bias
Random Forest (1-step*)	1.927	2.600	1.440	1.135
Gradient Boosting (1-step*)	1.999	2.704	1.494	1.287

Table 3. Tree-ensemble accuracy. *One-step forecasts see the most recent actual load (leakage), shown for contrast only.

As a fair recursive forecast the Random Forest scores MASE 1.78 — tied with seasonal naive — while its one-step version (1.44) only looks better because its lag-1 feature is fed last week's *actual* load. Feature importances confirm the point: lag_52 alone carries 77% of the Random Forest's weight, so the model mostly re-learns the annual cycle the benchmark already provides. The LSTM (Figure 10) posts the lowest score of all (weekly-aggregated MASE 0.879) but is a walk-forward one-step forecaster fed the true preceding 48 hours; its raw hourly RMSE (3.81 GW) is ordinary and aggregation to weekly flattens the headline — it is the neural analogue of the one-step tree leakage.

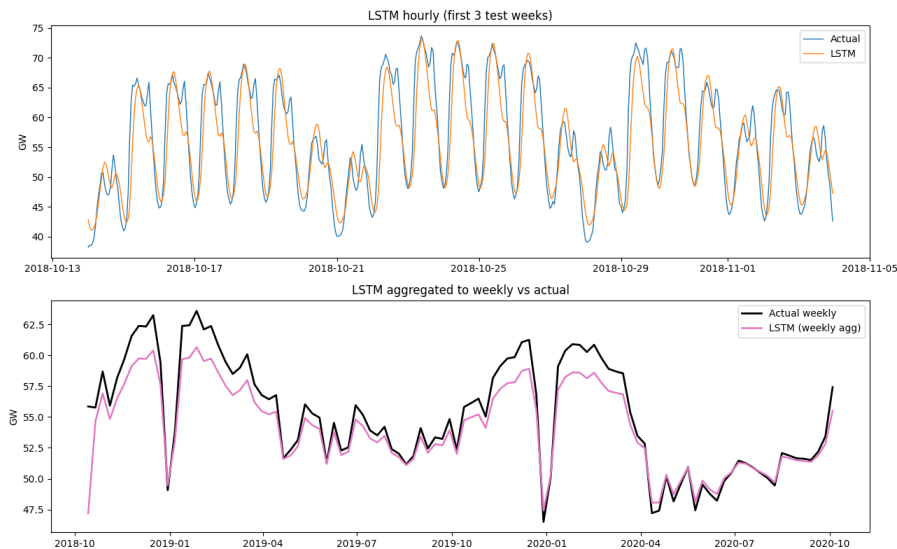


Figure 10. LSTM: hourly predictions for the first three test weeks (top) and the hourly forecast aggregated to weekly (bottom).

7. Analysis of Results

Model	MAE	RMSE	MASE	Bias	Forecast type
LSTM (hourly→weekly)	1.176	1.623	0.879	-1.050	1-step (leak)
Random Forest (1-step*)	1.927	2.600	1.440	1.135	1-step (leak)
Gradient Boosting (1-step*)	1.999	2.704	1.494	1.287	1-step (leak)
Seasonal naive	2.319	3.007	1.732	1.732	Multi-step
Random Forest (recursive)	2.383	3.018	1.781	1.274	Multi-step
Gradient Boosting (recursive)	2.531	3.317	1.891	1.561	Multi-step
SARIMAX (+temp, conditional)	2.851	3.589	2.131	2.557	Multi-step*
SARIMA	3.126	3.826	2.336	2.872	Multi-step
Naive	3.783	4.459	2.827	-0.882	Multi-step
Mean	3.789	4.397	2.831	0.481	Multi-step
Drift	4.340	5.118	3.243	1.007	Multi-step

Table 4. All models ranked by MASE. Seasonal naive (highlighted) is the best genuine multi-step forecast; rows above it consume recent actual load, and the SARIMAX row uses observed future temperature.

Q1. Compare all models against the seasonal naive; which, if any, provide a meaningful improvement?

Judged as genuine multi-step forecasts — the only fair basis — none beats the seasonal naive (MASE 1.73). The recursive Random Forest (1.78) matches it; SARIMAX (2.13), SARIMA (2.34) and the mean/naive/drift benchmarks are clearly worse. The three models that appear to win — the LSTM (0.88) and the one-step trees (1.44–1.49) — are not comparable, because each is fed recent actual load and therefore measures one-step tracking rather than two-year skill. No model provides a meaningful improvement over seasonal naive on this horizon.

Q2. How was data leakage avoided when creating temperature and lag features?

Every autoregressive feature was built with a strictly positive shift: load lags use $\text{shift}(k)$ with $k \geq 1$, and rolling means are computed on the once-shifted series $[\text{shift}(1).\text{rolling}(\dots)]$, so no feature can see the value it predicts. The temperature lag uses $\text{shift}(1)$ — the previous week. Most importantly, at prediction time the ensembles forecast *recursively*: each prediction is written back into the history and becomes the lag for the next step, so `lag_1` never reads a true future observation. Contemporaneous temperature is supplied as an exogenous input — a modelling assumption (conditional forecast, Q4), not lag leakage.

Q3. Justify the chosen differencing orders and seasonal period for SARIMAX.

The seasonal period $m = 52$ follows from weekly sampling of an annually cycling series (52 weeks per year), confirmed by periodic ACF spikes at lag 52 and the STL seasonal strength of 0.87. One seasonal difference ($D = 1$) removes that annual cycle — necessary because ADF/KPSS on the levels are blind to seasonal non-stationarity and understate it. One ordinary difference ($d = 1$) removes the residual low-frequency drift. Table 1 confirms the choice: after $d = 1$ and $D = 1(52)$ both tests report stationarity and the ACF/PACF (Figure 4) collapse to a few low-order spikes, which is why the AIC search settled on the low-order $(0,1,2)(0,1,1,52)$.

Q4. Do temperature and holiday covariates improve accuracy, and are they known at the forecast origin?

Temperature improves accuracy modestly (RMSE $3.83 \rightarrow 3.59$, MASE $2.34 \rightarrow 2.13$) and its positive coefficient confirms a real heating-driven signal. The crucial caveat is that temperature is not known at the forecast origin: the model was handed the observed test-period temperature, making it conditional/explanatory. An operational version would substitute a weather forecast and inherit its error, so the true gain is smaller than reported. Holiday covariates were not included here; unlike temperature they are deterministic and fully known in advance, making them a legitimate, leakage-free addition, though weekly averaging dilutes their single-day effect.

Q5. Compare the interpretability and complexity of SARIMAX, feature-based and neural-network models.

SARIMAX is the most interpretable and cheapest to fit: its differencing orders, MA terms and temperature coefficient all carry meaning and it yields calibrated prediction intervals, but it is linear. The tree ensembles are moderately interpretable through feature importances — which revealed `lag_52` carrying 77% of the Random Forest's weight — and capture non-linear interaction, at the cost of hand-built features and no native uncertainty. The LSTM is the most complex (scaling, windowing, many hyperparameters, GPU-scale training) and least interpretable, and despite that power did not out-perform the simpler models on a like-for-like basis.

Q6. Recommend one model for operational use, justified by accuracy, uncertainty, interpretability and maintenance.

For genuine multi-step operation I recommend SARIMA. On accuracy it sits in the same band as the recursive machine-learning models while being far simpler. On uncertainty it is the only candidate producing calibrated prediction intervals out of the box (Figure 7) — essential for reserve planning. On interpretability its structure and coefficients are transparent and auditable. On maintenance it is light: fast to refit, few dependencies, no GPU, unlike the LSTM. SARIMAX is preferable only where a trustworthy temperature forecast exists; where the real need is short-horizon, a walk-forward learner exploiting recent actuals is justified — but that is a different problem from the two-year forecast evaluated here. In all cases the seasonal naive should be retained as a monitoring benchmark.

8. Conclusions and Future Improvements

Compared like-for-like as genuine multi-step forecasts, no model meaningfully improves on the seasonal naive for two-year-ahead weekly German load. SARIMA fits in-sample well yet forecasts the anomalous horizon poorly; temperature helps only marginally and only under the unrealistic assumption of known future weather; the tree ensembles mostly re-learn the annual cycle the benchmark already captures; and the LSTM's apparent lead is an artefact of one-step walk-forward evaluation. Comparing the forecasts against the data collected after the modelling period, every model tracks the 2018–2019 seasonal wave closely but diverges sharply through 2020, confirming that the dominant error source is the unforecastable COVID-19 collapse rather than model misspecification.

Several improvements follow. **(i)** evaluate SARIMAX with forecast rather than observed temperature for an honest operational error; **(ii)** adopt rolling-origin backtesting with shorter horizons so scores are not dominated by one anomalous window; **(iii)** add deterministic calendar and public-holiday regressors; **(iv)** introduce intervention/anomaly handling (e.g. a lockdown indicator) so structural breaks are modelled, not ignored; and **(v)** train a direct multi-step (sequence-to-sequence) network on a GPU runtime with a longer look-back, evaluated on the same origin-fixed basis. Above all, the study argues for disciplined, leakage-free evaluation with the seasonal naive as the reference.

References

- Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015) *Time Series Analysis: Forecasting and Control*. 5th edn. Wiley.
- Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32.
- Cleveland, R.B. et al. (1990) 'STL: a seasonal-trend decomposition procedure based on loess', *J. Official Statistics*, 6(1), pp. 3–73.
- Dickey, D.A. and Fuller, W.A. (1979) 'Distribution of the estimators for autoregressive time series with a unit root', *JASA*, 74(366), pp. 427–431.
- Hochreiter, S. and Schmidhuber, J. (1997) 'Long short-term memory', *Neural Computation*, 9(8), pp. 1735–1780.
- Hyndman, R.J. and Athanasopoulos, G. (2021) *Forecasting: Principles and Practice*. 3rd edn. OTexts. <https://otexts.com/fpp3/>.
- Hyndman, R.J. and Koehler, A.B. (2006) 'Another look at measures of forecast accuracy', *Int. J. Forecasting*, 22(4), pp. 679–688.
- Kwiatkowski, D. et al. (1992) 'Testing the null hypothesis of stationarity against a unit root', *J. Econometrics*, 54(1–3), pp. 159–178.
- Open Power System Data (2020) Time series (2020-10-06 release). https://data.open-power-system-data.org/time_series/.
- Open-Meteo (2024) Historical Weather API. <https://open-meteo.com/>.